

Architecture — FindLostPuppy

High-Level Architectural Overview

FindLostPuppy is designed as a modern, responsive, privacy-first web platform for reporting and identifying lost dogs and collecting community sightings.

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``mermaid graph TD
  A[Client: React 18 + Vite] --> B[React Router DOM SPA]
  B --> C[Pages & Views]
  C --> C1[HomePage]
  C --> C2[DiscoveryPage]
  C --> C3[DogDetailPage]
  C --> C4[ReportWizardPage - 3 Chapters]
  C --> C5[DashboardPage]
  C --> D[Domain State & Services]
  D --> D1[AuthService - Email / Phone OTP / Session]
  D --> D2[StorageService - LocalStorage / IndexedDB Adapter]
  D --> D3[Location Privacy Module - GPS & Approximate Area]
  D2 --> E[Future Cloud Backend: Firebase / Cloud Run]
  F[CI/CD: GitHub Actions] --> G[Production Build: Vite]
  G --> H[GitHub Pages: jksurampudi5.github.io/findlostpuppy]`
```

Core Principles

- 1. Privacy Boundaries:** - Private data (exact street address, house/flat number, exact GPS meter coordinates, private phone numbers) is strictly partitioned from public representations. - Public representations display approximate regions (e.g. Near Benz Circle, Vijayawada or Near Bio-Diversity Park, Gachibowli, Hyderabad).
- 2. 2-Chapter Separation:** - Chapter 1: Owner Profile & Base Location. - Chapter 2: Pup Profile, Characteristics & Photos. - Chapter 3: Disappearance Event & Contact Preferences.
- 3. Offline-First & Extensibility:** - Built on a repository service abstraction (`storageService.ts`), allowing seamless substitution of local storage with Firebase or GraphQL backend APIs without modifying UI components.